LEARNING MODULE

22

Resampling Methods

A bootstrap reads an estimate's sampling error from resamples of the data; resampling single days ignores volatility clustering, and resampling blocks of days keeps part of it.

VOLUME IV · Statistical Inference and Econometrics

BUILDS ON LM4 · LM11 · LM13 · LM14 · LM15 · LM16 · LM17 · LM18 · LM19 · LM20 · LM21

LEADS TO LM23 · LM25 · LM26 · LM27 · LM30 · LM33 · LM34 · LM45

TIME ≈ 3 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	**1** explain the bootstrap principle, and compute a bootstrap standard error, a percentile interval and a basic interval, stating what BCa adds
☐	**2** calculate bootstrap intervals for a Sharpe ratio and a Kelly fraction, and compare them with the delta method's intervals
☐	**3** explain why resampling single days fails for dependent returns, and identify the statistics it misjudges
☐	**4** apply the moving-block and stationary bootstraps, state what circular blocks change, and explain what each scheme keeps and what it loses
☐	**5** choose a block length with the book's rule built on Politis and White's estimate, and check how much a result depends on it
☐	**6** compute block-bootstrap intervals for a variance ratio, a conditional correlation, a principal component's share and a window statistic, and choose between the percentile and the basic interval
☐	**7** perform a bootstrap test by resampling under the null hypothesis, and relate it to the interval at the same level
☐	**8** implement iid and block bootstraps in NumPy, check them against arch and SciPy, and report the seed, the resamples and the Monte Carlo error

LEARNING MODULE OVERVIEW

- The bootstrap replaces the unknown law F by the empirical $\hat{F}$ and reads an estimate's sampling error from resamples: a standard error (22.1) and percentile and basic intervals (22.2). More resamples shrink only the Monte Carlo error.
- On TR23's 5,849 days, iid resamples give the annual Sharpe ratio 0.81 the interval 0.41 to 1.21 and the Kelly fraction 4.28 the interval 2.2 to 6.5, close to LM20's delta method (0.40 to 1.22; 2.10 to 6.46).
- Resampling days destroys volatility clustering: the lag-1 autocorrelation of $|r|$ is 0.161, and none of 10,000 iid resamples reaches it.
- Blocks of days keep part of it. Stationary blocks of mean length 145 widen the volatility's interval from 18.3%–19.6% to 16.5%–21.5%, 3.9 times, but the Kelly fraction's only 1.2 times.
- Politis and White's estimate gives 3.1 days on returns and 144.6 on squared returns, and the book takes the larger. The volatility's standard error still grows at 400 days: a block keeps no dependence longer than itself.
- Block intervals answer questions of LM11, LM16 and LM18: the crash correlation lies below the normal model's (gap −0.094 to −0.033), and the first component's share was 30.1%–39.4% in 2017 and 49.6%–75.9% in 2020.
- A bootstrap test resamples a world where the null holds. Centring the 7% regime's returns gives a p-value of 0.048 ± 0.002 with blocks, beside LM21's normal 0.041.
- arch's block bootstraps draw the same indices as a few lines of NumPy from the same generator; SciPy's `bootstrap` resamples days only, interpolates its percentiles and defaults to BCa.

1 INTRODUCTION

LM20 gave each estimate a standard error by formula, for independent days and a smooth statistic. The **bootstrap** reads the error from the data instead. It recomputes the estimate on LM18's resamples. This module turns their spread into intervals and tests, and shows where resampling single days fails.

MOTIVATING CASE **How wide is a Kelly fraction?**

From 3 January 2003 to 8 October 2026 the TAIEX total-return index gave 5,849 daily returns, with mean 0.062505% and standard deviation 1.2084%. With no risk-free rate its Sharpe ratio is 0.81 a year and its volatility 18.96% a year, both by LM19's $\sqrt{A}$ convention. Its Kelly fraction $\hat{f} = \bar{R}/s^2$ is 4.28 (LM20).

Draw the days again at random, 10,000 times, and recompute each number. Of the resampled Kelly fractions, 95% lie between 2.2 and 6.5, like LM4's 2.7 to 8.6 for 0050's $\hat{\mu}/s^2$. Blocks of about 145 days, which keep calm and turbulent stretches together, widen the Kelly range only to 1.8 to 7.1.

Yet the volatility's range, 18.3% to 19.6% from single days, becomes 16.5% to 21.5% with blocks, almost four times as wide (Exhibit 1). Which range is an interval for TAIEX, and why do clusters matter so much more for one number?

Exhibit 1: Three estimates and their bootstrap intervals

Statistic	Estimate	iid days, 95%	Blocks of mean 145 days, 95%	Width ratio
Sharpe ratio, annualized	0.81	0.41 to 1.21	0.37 to 1.27	1.1
Kelly fraction $\hat{f}$	4.28	2.2 to 6.5	1.8 to 7.1	1.2
Volatility, annualized	18.96%	18.3% to 19.6%	16.5% to 21.5%	3.9

Note: TR23, $r_f = 0$, $A = 246.12$; the Sharpe ratio and the volatility are annualized by $\sqrt{A}$, a convention on simple returns (LM19). Estimates: LM19's and LM20's. Percentile intervals, $B = 10{,}000$: iid days (seed `[2, 20261011]`) and stationary blocks of mean length 145 (seed `[3, 20261011]`). Monte Carlo standard errors of the ends: at most 0.009, 0.04 and 0.03 points. Width ratios from unrounded ends. In-sample. Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026. Computations: `data/lm22_numbers.py`.

Sections 2 and 3 build intervals, Section 4 shows what resampling days misses, and Sections 5 and 6 resample blocks. Sections 7 to 9 answer earlier questions, test, and show the Python.

NOTE **Notation for this module**

TR23: closes 2 January 2003 → 8 October 2026, $T = 5{,}849$ daily simple returns R_t, log returns $r_t = \ln(1 + R_t)$, $A = 246.12$; 3,077 returns in the 7% limit regime (to 29 May 2015), 2,772 in the 10% regime. U28: Volume II's 28 TWSE sector price indices, without dividends, $T = 4{,}382$ returns from 2 December 2008. Moments divide by T; s_{u} (divisor $T - 1$) enters only the t-stat.

Local symbols: $\hat{\theta} = h(x_1, \ldots, x_T)$ an estimate of θ; θ_0 a value of θ under test; $\hat{\theta}_{\mathrm{boot},i}$ the statistic on resample i, with mean $\bar{\theta}_{\mathrm{boot}}$ and distribution $\hat{F}_{\mathrm{boot}}$ over the B resamples; $\mathbb{E}_{\hat{F}}$ an expectation under $\hat{F}$, (13.12); ℓ a block length or mean block length; $\widehat{\mathrm{SR}} = \bar{R}/s$ per day; $\hat{f} = \bar{R}/s^2$ and $\hat{f}^*$ the root of (4.5); $\hat{\rho}_1(|r|)$ the lag-1 autocorrelation (19.5) of $|r_t|$; α a significance level; G a growth factor; x_t, γ_0, ρ_k a daily series, its variance and its autocorrelations (Section 4). Indices: i resamples, t days, j ranks, blocks and a block's length in days.

> **NOTE** **Research hygiene**
>
> Every interval and test here resamples the sample it describes, so all are **in-sample**; those on TR23's full sample mix both limit regimes. Resampled paths, the widget and iid resamples read as independent days are a **model**. The 7% regime's test, Problem 11 and the lab's regime correlations are **stability checks**; LM11's shares and LM18's windows, as quoted, are **descriptive**. Nothing here is out-of-sample (LM34). Section 8 counts the tests.

2 THE BOOTSTRAP PRINCIPLE

- [ ] explain the bootstrap principle, and compute a bootstrap standard error, a percentile interval and a basic interval, stating what BCa adds

An estimate $\hat{\theta} = h(x_1, \ldots, x_T)$ varies from sample to sample. Its **sampling distribution** is its law when the T observations come from the unknown distribution F, and its standard error is that law's standard deviation. One sample shows one draw, and LM20 approximated the law by formulas such as (20.8).

The **plug-in principle** replaces F by $\hat{F}$, mass $1/T$ on each observation, (13.12) (Efron and Tibshirani 1993, ch. 4). A **resample** is T iid draws from $\hat{F}$ (LM18), and $\hat{\theta}_{\text{boot}}$ is the statistic on it. The **bootstrap principle**: $\hat{\theta}_{\text{boot}}$ varies under $\hat{F}$ roughly as $\hat{\theta}$ varies under F.

The T^T ordered resamples are equally likely, and the **ideal bootstrap** averages over all of them. For $T = 5{,}849$ that is impossible, so B random resamples approximate it, with LM18's Monte Carlo error.

> **DEFINITION** **Bootstrap standard error**
>
> Let $\hat{\theta}_{\text{boot},1}, \ldots, \hat{\theta}_{\text{boot},B}$ be the statistic on B resamples from the resampling model (iid days here, blocks from Section 5), and $\bar{\theta}_{\text{boot}}$ their mean. The **bootstrap standard error** is
>
> $$\widehat{\text{SE}}_{\text{boot}} = \left(\frac{1}{B} \sum_{i=1}^{B} \left(\hat{\theta}_{\text{boot},i} - \bar{\theta}_{\text{boot}} \right)^2 \right)^{1/2}. \tag{22.1}$$
>
> As $B \to \infty$ it tends to the ideal bootstrap's standard error, by the strong law (LM18).

Efron and Tibshirani (1993, §6.2) divide by $B - 1$ in their equation 6.6; the factor $\sqrt{(B-1)/B}$ is 0.99995 at $B = 10{,}000$.

EXAMPLE 1

The ideal bootstrap of five returns

Take LM18's returns +1.2%, −0.8%, +0.3%, −2.5% and +0.6% ($T = 5$). Find the ideal bootstrap standard errors of their mean and median, then estimate both from $B = 1{,}000$ resamples of `default_rng([1, 20261011])`.

SOLUTION

The mean is −0.24% and $s = 1.3032\%$. The $5^5 = 3{,}125$ ordered resamples are equally likely, so by (18.8) the resample mean has standard deviation $s/\sqrt{5} = 0.583\%$ exactly.

The resample median $\hat{F}^{-1}(\frac{1}{2})$ is the third smallest draw. It equals −2.5% or +1.2% with probability 181/3,125 each, −0.8% or +0.6% with 811/3,125 each, and +0.3% with 1,141/3,125, so its ideal standard error is 0.859%. Experiment 1 gives $0.59\% \pm 0.01\%$ and $0.87\% \pm 0.02\%$ (batch standard errors).

Intervals. Suppose that the error $\hat{\theta} - \theta$ under F has about the law of $\hat{\theta}_{\text{boot}} - \hat{\theta}$ under $\hat{F}$. Then $\hat{\theta} - \theta$ lies between the $\frac{\alpha}{2}$ and $1 - \frac{\alpha}{2}$ quantiles of $\hat{\theta}_{\text{boot}} - \hat{\theta}$ with probability about $1 - \alpha$. Solving for θ gives the basic interval.

DEFINITION **Percentile and basic intervals**

Let $\hat{F}^{-1}_{\text{boot}}(u) = \hat{\theta}_{\text{boot},(\lceil uB \rceil)}$ be the quantile function of the B resampled values (LM14). The $1 - \alpha$ **percentile interval** and **basic interval** are

$$\begin{aligned} &\big[\hat{F}^{-1}_{\text{boot}}(\tfrac{\alpha}{2}),\ \hat{F}^{-1}_{\text{boot}}(1 - \tfrac{\alpha}{2})\big], \\ &\big[2\hat{\theta} - \hat{F}^{-1}_{\text{boot}}(1 - \tfrac{\alpha}{2}),\ 2\hat{\theta} - \hat{F}^{-1}_{\text{boot}}(\tfrac{\alpha}{2})\big]. \end{aligned} \tag{22.2}$$

At $B = 10{,}000$ and $\alpha = 5\%$ the ends are the 250th and 9,750th resampled values.

Both intervals are approximate (Efron and Tibshirani 1993, ch. 13, for the percentile interval). Stated without proof: their coverage tends to $1 - \alpha$ for iid data and a smooth function of sample means with finite variances, such as $\bar{R}/s$. For $\bar{R}/s$ one mean is of R_t^2, whose variance needs a finite fourth moment; LM15's $t_{3.065}$ fit lacks it, so TR23's intervals rest on practice. Deep Dive (a) shows when the percentile interval is exact.

BCa adjusts the percentile interval for bias, measured by the share of resamples below $\hat{\theta}$, and for skewness, measured by a jackknife. A jackknife recomputes the statistic with each day left out once. BCa's adjustments are stated here without proof (Efron 1987). This book does not use the parametric bootstrap (Wasserman 2004, §9.11), subsampling (Lahiri 2003, §7.3), bootstrap-t intervals or the wild bootstrap.

EXAMPLE 2

An interval for TAIEX's Sharpe ratio

Resample TR23's days iid, $B = 10{,}000$ times with `default_rng([2, 20261011])`. Give the annual Sharpe ratio's bootstrap standard error and 95% intervals, and compare them with LM20's delta method.

SOLUTION

Experiment 2 draws `integers(0, T, size=(500, T))` twenty times. The standard error is 0.205 a year. The percentile and basic intervals both run from 0.41 to 1.21, since the values are nearly symmetric, and SciPy's BCa interval is 0.42 to 1.22.

LM20's delta method gives 0.207 with the sample's skewness and kurtosis, 0.205 for normal returns, and the interval 0.40 to 1.22. All agree because the daily ratio, 0.0517, is small: its standard error is close to $1/\sqrt{T} = 0.0131$ a day.

DEEP DIVE **(a) When the percentile interval is exact** optional · click to expand

Here ψ is a strictly increasing function and $\sigma_\psi > 0$ a constant. Suppose that $\psi(\hat\theta) - \psi(\theta)$ under F and $\psi(\hat\theta_{\text{boot}}) - \psi(\hat\theta)$ under $\hat F$ both follow $\mathcal{N}(0, \sigma_\psi^2)$. Since ψ is increasing, the u quantile of $\hat\theta_{\text{boot}}$ is $\psi^{-1}\big(\psi(\hat\theta) + \sigma_\psi \Phi^{-1}(u)\big)$. The ideal percentile interval therefore runs from

$$\psi^{-1}\big(\psi(\hat\theta) - \Phi^{-1}(1 - \tfrac{\alpha}{2})\,\sigma_\psi\big) \quad \text{to} \quad \psi^{-1}\big(\psi(\hat\theta) + \Phi^{-1}(1 - \tfrac{\alpha}{2})\,\sigma_\psi\big).$$

Also, $\psi(\theta)$ lies within $\Phi^{-1}(1 - \frac{\alpha}{2})\,\sigma_\psi$ of $\psi(\hat\theta)$ with probability exactly $1 - \alpha$. Applying ψ^{-1} gives an exact interval for θ, the same one, found without knowing ψ or σ_ψ. The basic interval is exact instead when $\hat\theta_{\text{boot}} - \hat\theta$ has the law of $\hat\theta - \theta$. ■

導讀 **自助重抽法：用樣本代替母體**

估計量（estimator）的抽樣分布（sampling distribution）要靠許多組樣本才看得到，我們卻只有一組。自助重抽法（bootstrap）用重抽樣（resampling）解決這件事：把經驗分布（empirical distribution）$\hat F$ 當成母體（population），從中放回抽取 T 天，重算統計量（statistic）B 次，以這些值的分布代替抽樣分布。它們的標準差（standard deviation）就是標準誤（standard error）的估計；$B = 10{,}000$ 時取第 250 小與第 9,750 小的值，就得到 95% 的百分位數信賴區間（percentile confidence interval），以 $2\hat\theta$ 減去這兩端，則是基本信賴區間（basic interval）。偏誤加速校正之自助重抽信賴區間（BCa）再修正偏誤（bias）與偏態（skewness），是 SciPy 的預設。要記得：B 越大只是模擬誤差（Monte Carlo error）越小，信賴區間（confidence interval）的寬度由樣本長度 T 與第 5 節的相依性（dependence）決定，與 B 無關。

KNOWLEDGE CHECK 1

Why can more resamples not bring Example 1's bootstrap standard error of the mean below 0.583%?

ANSWER

The number of resamples governs the Monte Carlo error only. As $B \to \infty$, (22.1) tends to the ideal value $s/\sqrt{5} = 0.583\%$, the sampling error of a mean of five returns under $\hat{F}$. Only a longer sample shrinks it.

3 INTERVALS FOR THE SHARPE RATIO AND THE KELLY FRACTION

- ☐ calculate bootstrap intervals for a Sharpe ratio and a Kelly fraction, and compare them with the delta method's intervals

The bootstrap needs no derivative. It recomputes the Kelly fraction $\hat{f} = \bar{R}/s^2$ (LM20) on each resample, and the exact optimum $\hat{f}^*$ by solving LM4's (4.5) on each one.

EXAMPLE 3

The Kelly fraction and the exact optimum

On experiment 2's resamples, find intervals for $\hat{f}$ and for $\hat{f}^*$, the root of (4.5) by Newton's method. Compare them with LM20's delta method, and repeat with experiment 3's blocks of mean length 145 (Section 5).

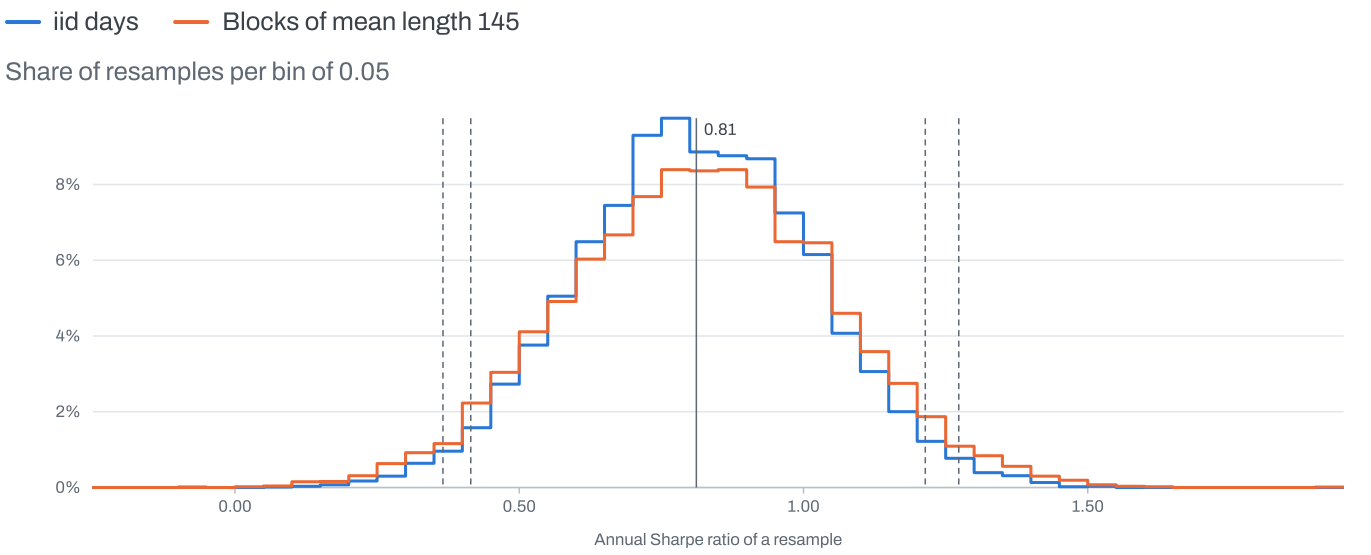
SOLUTION

For $\hat{f} = 4.28$ the iid resamples give a standard error of 1.10, the percentile interval 2.2 to 6.5, the basic interval 2.1 to 6.4 and SciPy's BCa interval 2.2 to 6.5. LM20's delta method gives 1.111 with plug-in skewness and kurtosis (1.085 for normal returns, 1.082 from the mean alone) and the interval 2.10 to 6.46.

The exact optimum $\hat{f}^* = 4.14$ (LM17) has standard error 1.03 and interval 2.1 to 6.1, lower and narrower. With blocks (`default_rng([3, 20261011])`) the intervals are 1.8 to 7.1 for $\hat{f}$ and 1.8 to 6.6 for $\hat{f}^*$ (Monte Carlo errors of every end at most 0.05).

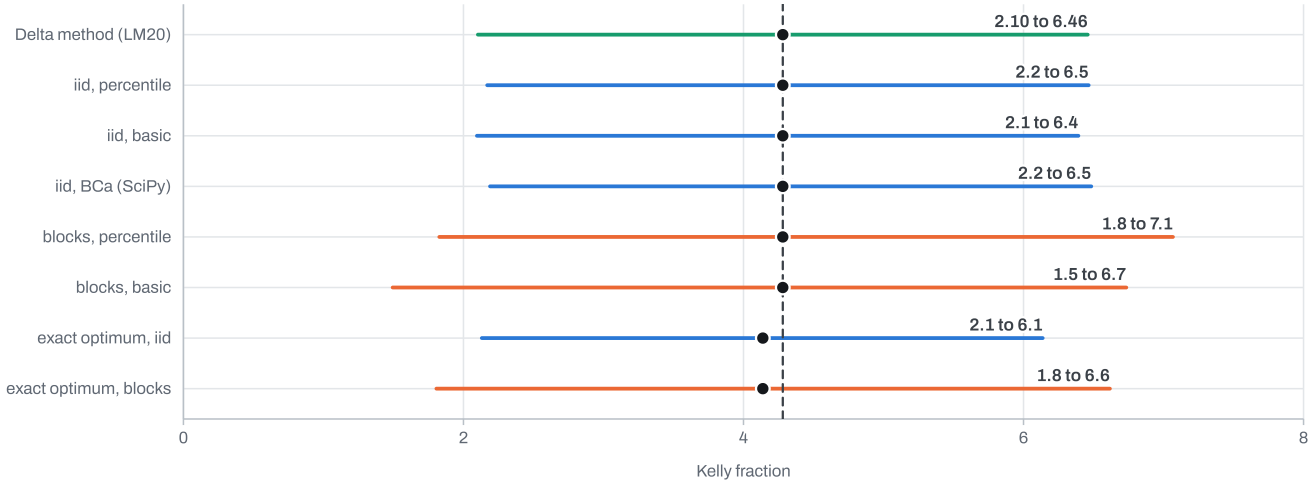
Exhibit 2 shows Example 2's resampled Sharpe ratios beside block resamples, and Exhibit 3 the Kelly intervals. The iid bootstrap agrees with the delta method because both treat the days as independent.

Exhibit 2: Resampled Sharpe ratios



Note: TR23's annual Sharpe ratio on 10,000 resamples each, share per bin of 0.05: iid days (experiment 2) and stationary blocks of mean length 145 (experiment 3). Solid line: the estimate 0.81. Dashed lines: the percentile ends, 0.41 and 1.21 (iid) and 0.37 and 1.27 (blocks). Standard errors: 0.205 and 0.233. In-sample. Source: as Exhibit 1.

Exhibit 3: The Kelly fraction's 95% intervals



Note: TR23, $B = 10{,}000$; iid rows from experiment 2, block rows from experiment 3. Points: $\hat{f} = 4.28$ in the first six rows, $\hat{f}^* = 4.14$ in the last two. Delta method (LM20's): 2.10 to 6.46. iid days: percentile 2.2 to 6.5, basic 2.1 to 6.4, BCa 2.2 to 6.5. Blocks: percentile 1.8 to 7.1, basic 1.5 to 6.7. Exact optimum: 2.1 to 6.1 (iid), 1.8 to 6.6 (blocks). In-sample. Source: as Exhibit 1.

The relative error. Most of the error of $\hat{f}$ is the error of $\bar{R}$. By (20.2) its relative error is $(s/\sqrt{T})/\bar{R} = 1/(\widehat{\mathrm{SR}}\sqrt{T})$, 25.3% here, and 1.96 such errors make about ±50%. The annual Sharpe ratio's interval is the daily one times $\sqrt{A}$, a convention on simple returns, exact only for log returns on uncorrelated days (LM19; Lo 2002).

LM7 used an illustrative $r_f = 1.5\%$ for 0050, while Volume IV's Sharpe ratios set $r_f = 0$: FinMind has no Taiwan rate series. Standard errors for dependent returns, the Probabilistic and Deflated Sharpe ratios and Ledoit and Wolf's (2008) test of two Sharpe ratios are LM33's.

PITFALL **An interval for the Kelly fraction is not a sizing rule**

The interval 2.2 to 6.5 says how far 4.28 could lie from the Kelly fraction behind TR23, if days were independent; it does not say which leverage to hold. Sizing from this distribution is LM45's subject, and a walk-forward test is LM34's.

KNOWLEDGE CHECK 2

TR23's iid percentile interval for $\hat{f}$ has a half-width of 50% of 4.28, and LM4's band for 0050's $\hat{\mu}/s^2$ one of 52% of 5.68. Why are the relative errors so close?

ANSWER

The relative error of $\bar{R}/s^2$ is about that of $\bar{R}$, $1/(\widehat{\mathrm{SR}}\sqrt{T})$: 25.3% here and 26% for 0050 (LM4). Both samples have $\widehat{\mathrm{SR}}\sqrt{T}$ near 4 (3.96 and 3.79), and 1.96 relative errors of about 25% make about ±50%.

4 WHEN RESAMPLING DAYS FAILS

- [ ] explain why resampling single days fails for dependent returns, and identify the statistics it misjudges

An iid resample's days are independent by construction, so a statistic whose error depends on the order of the days gets the wrong error. The variance of a mean shows which statistics are affected.

Let x_t be a weakly stationary daily series, the condition of (19.7). By (19.7) with $m = T$, $\sum_t x_t$ has variance $T\gamma_0 + 2\sum_{k=1}^{T-1}(T-k)\gamma_k$. Dividing by T^2 gives

$$\operatorname{Var}\bar{x} = \frac{\gamma_0}{T}\Big(1 + 2\sum_{k=1}^{T-1}\big(1 - \tfrac{k}{T}\big)\rho_k\Big).$$

An iid resample makes every ρ_k zero in expectation, so it reports about γ_0/T. For the mean of R the autocorrelations are small (LM19), and the iid error is close. But s^2 is the mean of $(R_t - \bar{R})^2$, and squares are strongly autocorrelated: volatility clusters. So the iid bootstrap understates the volatility's error and cannot measure a statistic of dependence at all.

EXAMPLE 4

What resampling days cannot see

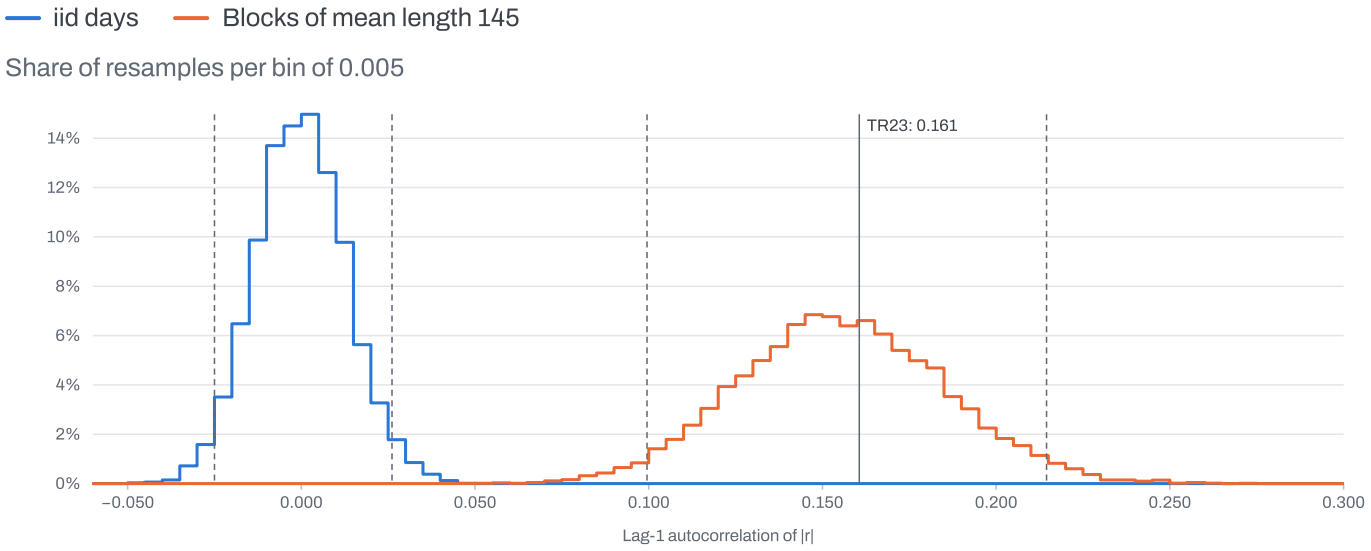
TR23's lag-1 autocorrelation of $|r_t|$ is $\hat{\rho}_1(|r|) = 0.161$ (LM19; LM17's pair correlation prints the same). TR23's log returns have the variance ratio $\mathrm{VR}(20) = 1.036$ (LM19). Recompute both on experiments 2 and 3.

SOLUTION

On the iid resamples $\hat{\rho}_1(|r|)$ averages −0.0001, and 95% of the values lie between −0.025 and 0.026: none of 10,000 reaches 0.161 (Exhibit 4). On the block resamples it averages 0.156, with percentile interval 0.10 to 0.21.

For VR(20) the iid resamples average 0.997, with 95% of them between 0.88 and 1.13, centred on independent days instead of on 1.036. The blocks average 1.037, with 95% between 0.92 and 1.16. The standard error barely moves, 0.065 against 0.061, but only the blocks centre the interval on the data. Monte Carlo standard errors of the interval ends are at most 0.0007 ($\hat{\rho}_1$) and 0.002 (VR(20)).

Exhibit 4: What resampling days destroys



Note: $\hat{\rho}_1(|r|)$ (LM19's convention) on TR23's log returns and on 10,000 resamples each, share per bin of 0.005. Resamples: iid days (experiment 2) and stationary blocks of mean length 145 (experiment 3). Solid line: the estimate 0.161. Dashed lines: the percentile ends, −0.025 and 0.026 (iid) and 0.10 and 0.21 (blocks). Model (iid days) and in-sample (blocks). Source: as Exhibit 1.

For large T, the product $T \operatorname{Var} \bar{x}$ approaches the long-run variance, which LM23 estimates directly (HAC); the block bootstrap estimates it by resampling.

KNOWLEDGE CHECK 3

Why do iid resamples give VR(20) near 1, whatever the data?

ANSWER

Resampled days are independent draws from $\hat{F}$, so their autocorrelations vanish in expectation, and by (19.7) a 20-day sum has variance close to $20s_r^2$. The resamples average 0.997, near LM19's iid factor of (19.8), $(1 - 20/T)/(1 - 1/T) = 0.997$.

5 BLOCK BOOTSTRAPS

- [ ] apply the moving-block and stationary bootstraps, state what circular blocks change, and explain what each scheme keeps and what it loses

A block bootstrap resamples stretches of consecutive days, so the dependence inside each stretch survives. **Moving blocks** (Künsch 1989) draw $\lceil T/\ell \rceil$ of the $T - \ell + 1$ stretches of ℓ days with replacement, join them and cut the result to T days. Days near the sample's ends lie in fewer blocks and are drawn less often (Deep Dive (b)).

DEFINITION **Stationary bootstrap**

The **stationary bootstrap** with mean block length ℓ (Politis and Romano 1994) fills a resample position by position. Its first day is uniform on the T days. At each later position, with probability $1/\ell$ a new block starts on a uniform day; otherwise the next day follows, with day T followed by day 1. Hence

$$\begin{aligned} &\Pr(\text{a block lasts at least } j \text{ days}) \\ &\quad = \left(1 - \frac{1}{\ell}\right)^{j-1}, \end{aligned} \tag{22.3}$$

the mean block length is ℓ, and every position is uniform on the T days (Deep Dive (b)).

Circular blocks are moving blocks on a circle: day T is followed by day 1, so every day is equally likely (Lahiri 2003, §2.7.1). Stationarity is defined in LM25, and volatility that depends on yesterday's is modelled in LM26.

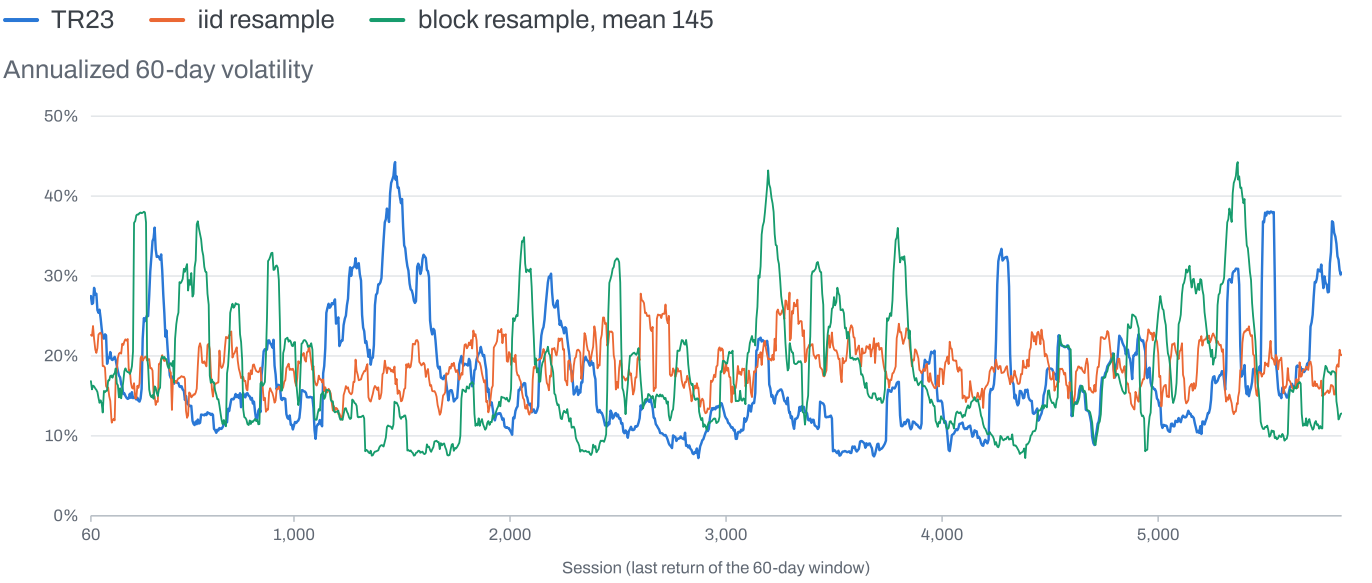
Stated without proof (Lahiri 2003, Theorem 3.1): let a stationary series have finite moments of some order above 2, and let its dependence die out fast enough (strong mixing). Then moving and circular blocks estimate the variance of its mean consistently if $\ell \to \infty$ and $\ell/T \to 0$. For the stationary bootstrap, Theorem 3.3 asks more: moments of some order above 4, faster-dying dependence, and $\ell/\sqrt{T} \to 0$.

LM15's Student t fit to TR23 has 3.065 degrees of freedom, so its moments exist only below order 3.065. That meets Theorem 3.1's moment condition for the mean of R_t but not Theorem 3.3's, so the stationary bootstrap of the mean (Example 10) rests on practice too.

The volatility, Sharpe ratio and Kelly fraction, $\mathrm{VR}(20)$, $\hat{\rho}_1(|r|)$, the windows' spread and Section 7's correlations and component shares average squares or products of returns. For these, Theorem 3.1 needs moments above order 4, so their intervals rest on the method's practice; for the first three, the three block schemes give similar widths (Exhibit 6).

In Exhibit 5 the iid path turns TR23's calm and turbulent stretches into an even 12% to 28% a year, while the block path copies whole stretches, extremes included.

Exhibit 5: One path of each kind



Note: the annualized standard deviation of each window of 60 returns (divisor 60), at the window's last session, for TR23 and two resamples of experiment 7. TR23 ranges from 7.2% (window ending 29 July 2014) to 44.3% (ending 2 December 2008). The iid path ranges from 11.6% to 27.9%, the block path (mean length 145) from 7.2% to 44.3%; there a session is a position. Every fifth session is plotted, plus the last and each path's extremes. Model. Source: as Exhibit 1.

EXAMPLE 5

TAIEX's volatility with blocks

Compare the bootstrap standard errors of the annual volatility, 18.96%, under iid days and under stationary blocks of mean length 145, with LM20's iid formula.

SOLUTION

LM20's formula $\mathrm{SE}(s) \approx s\sqrt{(\mathrm{Kurt}-1)/(4T)}$, with LM14's Kurt 8.27, gives 0.33 points a year, and so does the iid bootstrap. The stationary blocks give 1.30 points, 3.9 times as much: a relative error of about 7% instead of under 2%, and the interval 16.5% to 21.5%. Moving and circular blocks give 1.12 points (Exhibit 6).

Exhibit 6: Four resampling schemes

Scheme	Volatility SE (points)	Volatility, 95%	Kelly SE	Kelly, 95%	Sharpe SE
iid days	0.33	18.3% to 19.6%	1.10	2.2 to 6.5	0.205
Moving blocks	1.12	16.5% to 21.0%	1.45	1.6 to 7.2	0.247
Circular blocks	1.12	16.7% to 21.1%	1.41	1.7 to 7.1	0.244
Stationary bootstrap	1.30	16.5% to 21.5%	1.34	1.8 to 7.1	0.233

Note: TR23; annual volatility and Sharpe ratio, $\hat{f}$; blocks of 145 days (a mean for the stationary bootstrap); $B = 10{,}000$ each; percentile intervals. Experiments 2 (iid), 5 (moving: 41 starts among 5,705 blocks), 6 (circular) and 3 (stationary). Monte Carlo standard errors (batch) of the ends: at most 0.04 points (volatility) and 0.05 (Kelly). In-sample. Source: as Exhibit 1.

The printed intervals use the stationary bootstrap, whose resampled series is stationary given the data (Politis and Romano 1994).

DEEP DIVE **(b) Every day is equally likely at every position** optional · click to expand

In a stationary or circular resample, suppose that the day at one position is uniform on the T days. The next position holds a new uniform draw or the day after (day T followed by day 1), chosen independently of the day. A uniform day shifted one place around the circle is still uniform, so the next position is uniform too.

By induction from the first, every position is uniform, with mean $\bar{R}$ under $\hat{F}$, so $\mathbb{E}_{\hat{F}} \bar{R}_{\text{boot}} = \bar{R}$ for every ℓ.

Moving blocks have no wrap. Day t lies in $\min(t, \ell, T - \ell + 1, T - t + 1)$ of the $T - \ell + 1$ blocks, so the first and last days are drawn less often. ■

PITFALL **A block keeps no dependence longer than itself**

Blocks of mean length 145 days keep clustering that fades within months. Dependence over years, such as LM19's variance ratios below 1 beyond 340 days, is cut at every block's end. So the volatility's standard error still rises at 400 days (Section 6), and almost no block (about 1 in 5,000) holds a five-year window (Section 7).

導讀 **區塊自助重抽法：保留波動群聚**

一天一天地重抽，會把波動群聚現象（volatility clustering）完全打散：資料中 $|r_t|$ 的一階自我相關（autocorrelation）是 0.161，獨立同分布（iid）重抽的結果卻都在 0 附近。區塊自助重抽法（block bootstrap）改為整段抽取連續的交易日：移動區塊（moving block）從所有長度為 ℓ 的區段中抽取，循環區塊（circular block）把樣本頭尾相接，平穩自助重抽法（stationary bootstrap）則讓區塊長度（block length）隨機，平均為 ℓ。整段抽取保留了平靜與動盪的期間，所以波動率（volatility）的信賴區間（confidence interval）寬度變成將近原來的四倍，凱利比例（Kelly fraction）的信賴區間卻只寬一點，因為它的誤差主要來自平均報酬（mean return）。比區塊更長的相依性（dependence），仍然會在區塊的接縫處被切斷。

KNOWLEDGE CHECK 4

How many moving blocks of 145 days does a resample of 5,849 days need, and how many drawn days are cut?

ANSWER

It needs $\lceil 5{,}849/145 \rceil = 41$ blocks, which hold $41 \times 145 = 5{,}945$ days, so 96 days are cut from the last block.

6 CHOOSING THE BLOCK LENGTH

- [ ] choose a block length with the book's rule built on Politis and White's estimate, and check how much a result depends on it

No block length suits every statistic. Politis and White (2004), as corrected by Patton, Politis and White (2009), estimate the length that makes the block bootstrap's variance of a **mean** most accurate. Their target is the mean of the series they are given (stated without proof). arch's `optimal_block_length` computes it for stationary and circular blocks, capped at $\lceil \min(3\sqrt{T}, T/3) \rceil$.

EXAMPLE 6

A block length for TR23

Apply `optimal_block_length` to R, to $(R - \bar{R})^2$ and to $(r - \bar{r})^2$, and choose one length for TR23.

SOLUTION

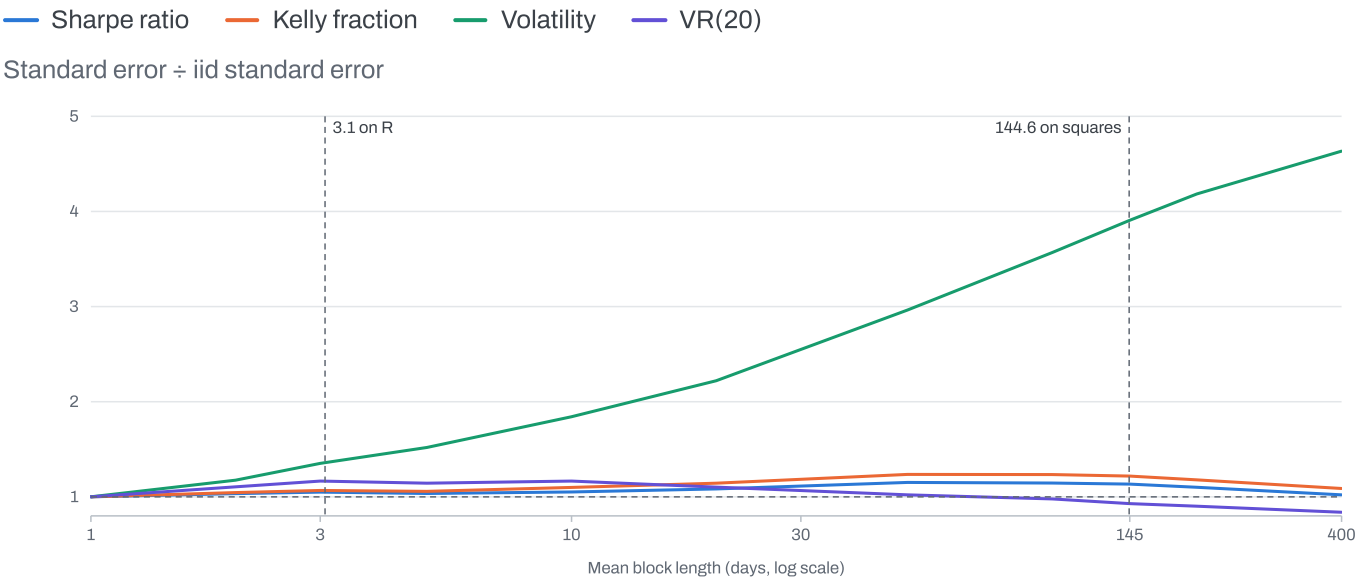
On R the estimate is 3.07 days for the stationary bootstrap (3.51 for circular blocks), since returns are nearly uncorrelated. On $(R - \bar{R})^2$ it is 144.59 (165.52), and on $(r - \bar{r})^2$ 144.02; the cap is 230.

The **book's rule** takes one length per sample and family of statistics: the largest stationary estimate among the squared demeaned simple-return series the statistic involves, rounded. For TR23 that is 145, for every statistic on R or r, $\mathrm{VR}(20)$ and $\hat{\rho}_1(|r|)$ included. It is a rule, not the right length.

The rule is conservative for statistics driven by the mean. In Exhibit 7 the standard errors of the Sharpe ratio and of the Kelly fraction grow by only 8% and 14% from $\ell = 3$ to 145. Their ratios to the iid error peak at 1.15 and 1.24, between $\ell = 50$ and 100.

The volatility's ratio is 1.18 at $\ell = 2$, 3.91 at 145 and 4.63 at 400, still rising. Dependence in the squares, from clustering or from shifts between volatility regimes, may last longer than any block a 23-year sample allows. For $\mathrm{VR}(20)$ the ratio is 0.93 at 145: blocks move its centre, not its width (Example 4).

Exhibit 7: Standard error against block length



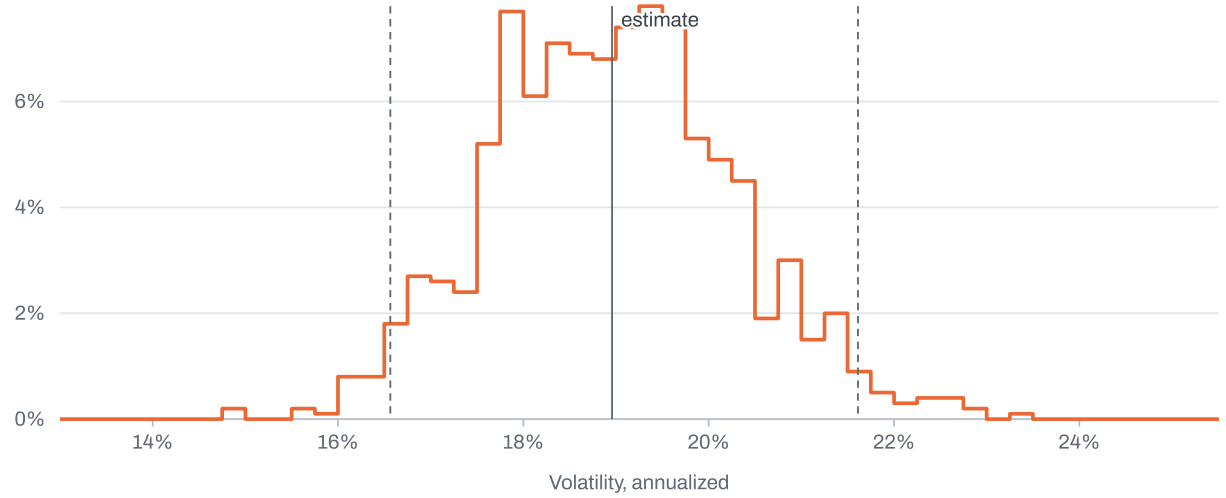
Note: TR23; the stationary bootstrap's standard error at mean block length ℓ over the iid one; logarithmic axis for ℓ. Points at 1 (iid) and 145: experiments 2 and 3. Others: experiment 4, $B = 2{,}000$ at each length, in increasing order. Dashed lines: Politis and White's 3.1 days on R and 144.6 on $(R - \bar{R})^2$. Monte Carlo standard error of a ratio: at most 2.2%, about 1% at 145. In-sample. Source: as Exhibit 1.

The widget shows the 1,000 resampled values of five statistics at each length of Exhibit 7, in three replicates of its own (experiment 24).

Block-bootstrap explorer

INTERACTIVE

Share of the 1,000 resamples per bin



Estimate on TR23

18.96%

Bootstrap standard error (divisor 1,000)

1.30 points

95% percentile interval

16.56% to 21.61%

95% basic interval

16.30% to 21.35%

Normal interval, estimate ± 1.96 standard errors

16.41% to 21.50%

Mean of the resamples

18.96%

Scheme · replicate

stationary blocks of mean length 145 · 1 of 3

Each histogram holds 1,000 resamples of TR23's daily returns, drawn in NumPy by one generator, default_rng([24, 20261011]) (experiment 24), 3 replicates at each length; length 1 resamples single days, the others use the stationary bootstrap. Nothing is drawn in the browser. Solid line: the estimate on TR23; dashed lines: the percentile ends. The replicates differ by Monte Carlo error, larger than in the text's 10,000 resamples. In-sample; the text quotes no value from this figure.

KNOWLEDGE CHECK 5

Politis and White give 3.1 days on R and 144.6 on squared returns. Why does the Sharpe ratio's interval hardly depend on which is used?

ANSWER

Its error is almost all the error of $\bar{R}$, which depends on the small autocorrelations of R, the series that the 3.1 days describe. In Exhibit 7 its standard error is 1.05 times the iid one at $\ell = 3$ and 1.13 times at 145.

7 INTERVALS THE EARLIER MODULES LEFT OPEN

- [ ] compute block-bootstrap intervals for a variance ratio, a conditional correlation, a principal component's share and a window statistic, and choose between the percentile and the basic interval

Exhibit 8 gives estimates of earlier modules, whose error has no formula, block-bootstrap intervals by the book's rule.

Exhibit 8: Block-bootstrap intervals for earlier estimates

Estimate (sample)	Value	ℓ	Bootstrap SE	95% percentile interval
Correlation on 223 big-move days (U28 pair)	0.7884	130	0.022	0.75 to 0.84
Its gap to (16.10)'s 0.8598 (U28 pair)	−0.0714	130	0.016	−0.094 to −0.033
First component's share, 2017 (U28)	34.5%	7	2.4 points	30.1% to 39.4%
First component's share, 2020 (U28)	66.6%	18	7.1 points	49.6% to 75.9% (basic 57.3% to 83.7%)
Five-year windows under water (TR23)	4.14%	145	9.9 points	0% to 33%
Standard deviation of their $\ln G$ (TR23)	0.283	145	0.124	0.20 to 0.68
VR(20) (TR23)	1.036	145	0.061	0.92 to 1.16

Note: stationary bootstrap at the book's block length for each sample; $B = 10{,}000$; experiments 9, 9, 11, 13, 14, 14 and 3. Monte Carlo standard errors (batch) of the ends: at most 0.6 points for the windows' share, 0.16 points for the component shares and 0.005 elsewhere. U28 rows are price indices without dividends; the pair is Semiconductor and FinancialInsurance. In-sample. Sources: FinMind `TaiwanStockPrice` (sector indices, retrieved 11 October 2026, closes repaired as in LM16) and `TaiwanStockTotalReturnIndex` (TAIEX, retrieved 18 October 2026).

EXAMPLE 7

Is the crash correlation below the model's by chance?

LM16 found that Semiconductor and FinancialInsurance correlate 0.788 on the 223 days with $|R_t - \bar{R}| \geq 2s$ for Semiconductor, against 0.860 by (16.10) for a normal pair with the full-sample $\hat{\rho} = 0.5749$. Give the gap a block-bootstrap interval.

SOLUTION

To four decimals the data give 0.7884, and (16.10) with $\hat{\rho} = 0.57486$ and $\mathrm{Var}(Z \mid |Z| \geq 2) = 5.7464$ gives 0.8598: a gap of −0.0714. Each resample re-standardizes Semiconductor, reselects its big-move days (182 to 255 in the block resamples below) and recomputes both correlations.

Politis and White give 53.3 days on Semiconductor's squares and 130.0 on FinancialInsurance's, so $\ell = 130$. Experiment 9 gives the gap a standard error of 0.016 and the interval −0.094 to −0.033; none of the 10,000 gaps reaches 0. iid days (experiment 8) give −0.102 to −0.037. By LM21's duality the crash correlation is below the normal model's at the 5% level.

EXAMPLE 8

The first component's share from about 250 returns

LM11's first principal component carried $\lambda_1/N = 34.5\%$ of the 28 standardized sectors' variance in 2017 and 66.6% in 2020, (11.6). How much of the gap could be sampling error?

SOLUTION

In 2017 (246 returns) the largest Politis–White estimate over the 28 sectors' squares is 6.79, so $\ell = 7$. Experiment 11 gives a standard error of 2.4 points and the interval 30.1% to 39.4% (iid days, experiment 10: 30.7% to 38.9%).

In 2020 (245 returns) the estimate is 18.38, so $\ell = 18$. Experiment 13 gives a standard error of 7.1 points and the percentile interval 49.6% to 75.9% (iid, experiment 12: 58.6% to 72.8%). The resampled shares average about 1.7 points below the 2020 share (a bias), with a long left tail. The basic interval, from the ends to two decimals, 49.55% and 75.92%, moves up to 57.3% to 83.7%. The two years' percentile intervals do not overlap.

Percentile or basic. The basic interval reflects the resampled values about $\hat{\theta}$, so it corrects a bias but can leave the statistic's range. The percentile interval stays inside the range but inherits the bias. A bounded statistic near its bound needs the percentile interval.

EXAMPLE 9

The error of LM18's windows

Of LM18's 4,619 windows of $m = 1{,}231$ sessions, 191 ended under water, 4.14%. Their $\ln G$ has standard deviation 0.283 (divisor 4,619), against $\sqrt{m}\, s_r = 0.425$ for independent five-year paths (LM18). Give both statistics block-bootstrap intervals.

SOLUTION

Each stationary resample of TR23's log returns ($\ell = 145$, experiment 14) has its own 4,619 windows. The share under water has standard error 9.9 points and interval 0% to 33%; 3,248 of 10,000 resamples have no window under water. The basic interval, −25% to 8%, leaves the range of a share. iid days (experiment 15) give a standard error of 8.4 points.

The standard deviation of $\ln G$ has standard error 0.124, interval 0.20 to 0.68 and resampled mean 0.386 (iid days: 0.19 to 0.58, mean 0.357). Overlapping windows of one 23-year path understate the variance even for independent days, by about m/T (LM19's $\mathrm{VR}(m)$), 24% about their own mean. A mean of standard deviations also lies below the square root of the mean variance. So 0.357, not 0.425, is the benchmark.

Blocks keep the swings within months that lift the resampled mean to 0.386 (LM19's variance ratios above 1 below 340 days). But they cut the swings of years that hold TR23's windows at 0.283.

Example 4's $\mathrm{VR}(20)$ has the block interval 0.92 to 1.16, which includes 1. LM25 tests a variance ratio with its own robust standard error.

導讀 **百分位數信賴區間與基本信賴區間**

重抽樣（resampling）的分布（distribution）若有偏誤（bias）或偏態（skewness），百分位數信賴區間（percentile confidence interval）與基本信賴區間（basic interval）就會分開。基本信賴區間把重抽樣分布以估計值（estimate）為中心翻轉，能修正偏誤，卻可能超出統計量（statistic）的範圍：五年視窗（window）虧損比例的基本信賴區間下界約是 −25%，但比例不可能為負。百分位數信賴區間一定落在範圍內，卻會繼承偏誤：2020 年第一主成分（first principal component）占比的重抽樣平均，比估計值低約 1.7 個百分點。變異數比（variance ratio）$\mathrm{VR}(20)$ 以區塊自助重抽法（block bootstrap）算出的信賴區間是 0.92 到 1.16，包含 1。這些信賴區間都只描述已觀察到的樣本，不是預測。

KNOWLEDGE CHECK 6

The 2017 and 2020 intervals of Example 8 do not overlap. Does that show that the first component's share changed?

ANSWER

It is strong evidence of a change between those two years. But the two years were picked after looking, as the lowest and highest of LM11's 18 calendar years. The comparison is the best of a search, and LM21's count of tests applies.

8 BOOTSTRAP TESTS

- [ ] perform a bootstrap test by resampling under the null hypothesis, and relate it to the interval at the same level

A test needs the statistic's law when the null hypothesis holds (LM21). A bootstrap test builds that law by resampling a version of the data in which the null holds: for $\mu = 0$, the centred returns $R_t - \bar{R}$.

DEFINITION **Bootstrap p-value of a mean**

Draw B resamples of the centred returns $R_t - \bar{R}$ under a stated scheme, and compute each one's t-stat (21.1), $\text{t-stat}_{\text{boot},i}$. The two-sided **bootstrap p-value** is

$$\frac{1}{B}\,\#\big\{i : |\text{t-stat}_{\text{boot},i}| \geq |\text{t-stat}|\big\}. \tag{22.4}$$

Its Monte Carlo standard error is $\sqrt{\text{p-value} \times (1 - \text{p-value})/B}$.

EXAMPLE 10

A bootstrap test of the mean

In TR23's 7% regime (3,077 returns), LM21's t-stat is 2.05, with a two-sided normal p-value of 0.041. Test $\mu = 0$ at the 5% level with (22.4), iid and with stationary blocks; then test the full sample.

SOLUTION

Resampling $R_t - \bar{R}$ iid (experiment 16), 408 of 10,000 resampled $|\text{t-stat}|$ reach 2.05: a bootstrap p-value of 0.041 ± 0.002, as LM21 found. With stationary blocks of mean length 108, the regime's own rule (experiment 17), 477 reach it: 0.048 ± 0.002. The blocks barely widen the null law, whose standard deviation is 1.03 against 1.01 for iid days.

Both p-values are below 5%, so both reject, though the block one is close (Problem 7). For the full sample LM21 gives the t-stat 3.96 and the normal p-value 7.6×10^{-5}. With blocks of 145 (experiment 18) none of 10,000 resampled $|\text{t-stat}|$ reaches 3.96. That makes (22.4) zero, but zero hits only bound the p-value: below about $3/B = 0.0003$ at the 95% level. The decision at the 5% level stands.

Independence as a null. Example 4's iid resamples are draws under the null that days are independent. None reaches 0.161, which lies 12 standard errors (0.013) above 0, so independence of $|r_t|$ is rejected. LM21 built its calendar nulls by shuffling the dates, a resampling without replacement that imposes "no calendar effect" (*Model*); here centring imposes "no mean".

Tests and intervals. A two-sided test at level α rejects $\theta = \theta_0$ about when the $1 - \alpha$ interval (22.2) excludes θ_0, as in Example 7. They agree only approximately: the test resamples under the null, the

interval under the data.

How many tests. This module ran four tests: three on TR23 (the mean in the 7% regime and in the full sample, and independence) and one on U28. Each null counts once, whatever the scheme, and the U28 test is the crash gap's interval read by duality. The 7% regime is one of two, and Example 8 compared years chosen after looking (LM21). Testing many strategies with the bootstrap (the reality check) is LM34's.

導讀　自助重抽檢定：先造一個虛無假設成立的世界

假設檢定（hypothesis testing）需要統計量（statistic）在虛無假設（null hypothesis）成立時的分布（distribution）。自助重抽檢定（bootstrap test）先把資料改成虛無假設成立的樣子，例如把報酬減去平均數（mean），使平均報酬（mean return）為零，再重抽樣（resampling）、重算 t 統計量（t-statistic）。p值（p-value）就是重抽樣的 $|\text{t-stat}|$ 至少和實際值一樣大的比例。若直接重抽原始報酬，得到的是對立假設（alternative hypothesis）下的分布，t 統計量的中心在 3.96 附近，不能當檢定的基準。漲跌幅限制（price limit）為 7% 的時期以區塊自助重抽法（block bootstrap）算出的 p值是 0.048，在 5% 的顯著水準（significance level）下仍拒絕虛無假設，但很接近界線。

KNOWLEDGE CHECK 7

Why does Example 10's full-sample test resample $R_t - \bar{R}$ and not R_t?

ANSWER

To impose the null $\mu = 0$. iid resamples of R_t itself (experiment 19) give t-stats with mean 3.96, and half of them lie above the observed 3.96. That is the law under the alternative, useless as a null.

9 RESAMPLING IN PYTHON

- [ ] implement iid and block bootstraps in NumPy, check them against arch and SciPy, and report the seed, the resamples and the Monte Carlo error

The function below draws the stationary bootstrap in arch's order, T candidate start days and then T uniforms, and checks itself against arch (experiment 20). Here `R` holds TR23's daily simple returns:

PYTHON

```python
import numpy as np
from arch.bootstrap import StationaryBootstrap


def stationary_idx(rng, T, ell):
    """Day indices of one stationary resample, drawn in arch's order."""
    start = rng.integers(T, size=T)        # a candidate start day per position
    new = rng.random(T) <= 1 / ell         # a new block starts here
    new[0] = True
    first = np.flatnonzero(new)            # positions where blocks start
    blk = np.cumsum(new) - 1               # the block of each position
    step = np.arange(T) - first[blk]       # steps since its block started
    return (start[first][blk] + step) % T  # day T is followed by day 1


g1 = np.random.default_rng([20, 20261011])      # experiment 20: twins
g2 = np.random.default_rng([20, 20261011])
bs = StationaryBootstrap(145, R, seed=g1)
for _ in bs.bootstrap(3):
    print((bs.index == stationary_idx(g2, len(R), 145)).all())   # True
```

arch's `CircularBlockBootstrap` and `MovingBlockBootstrap` match their NumPy twins too. `optimal_block_length(x)` returns a pandas DataFrame with the columns `stationary` and `circular`, but its docstring calls them `b_sb` and `b_cb`. Resampled portfolios are LM30's.

`scipy.stats.bootstrap` resamples days only and differs from this book in three places. Its interval ends interpolate (`stats.quantile`, linear): 0.4157 against the book's 0.4147 for Example 2. Its `standard_error` divides by $B - 1$: 0.20539 against 0.20538. Its default is BCa with 9,999 resamples. To draw this book's resamples, pass `batch=500` and the generator as the keyword `rng`; `method="percentile"` selects the percentile interval:

PYTHON

```python
from scipy import stats

A = 246.12


def sharpe(x, axis=-1):                    # annual Sharpe ratio, r_f = 0
    return np.sqrt(A) * x.mean(axis=axis) / x.std(axis=axis)


res = stats.bootstrap((R,), sharpe, n_resamples=10_000, batch=500,
                      method="percentile",
                      rng=np.random.default_rng([2, 20261011]))
print(res.confidence_interval)             # low=0.4157... high=1.2142...
```

IN PYTHON **Companion lab**

The Polars-first notebook `LM22_lab.ipynb` rebuilds the module's exhibits and computational examples from TR23's closes and the repaired sector closes. It solves the computational practice problems and ends with block-bootstrap intervals for LM16's pair correlation in each limit regime, a stability check.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 8

You pass arch `seed=5`, and a colleague passes the generator `default_rng([5, 20261011])`. Do you draw the same resamples?

> **ANSWER**
>
> No. arch turns an integer seed into `np.random.default_rng(5)`, another stream. Pass the experiment's generator itself.

SUMMARY

- The bootstrap replaces F by $\hat{F}$: (22.1) and (22.2) come from resamples, and more resamples shrink only the Monte Carlo error.

- The iid bootstrap agrees with LM20's delta method: 0.41 to 1.21 for the Sharpe ratio, 2.2 to 6.5 for the Kelly fraction.
- Resampling days destroys dependence and understates the error of statistics of squares.
- Blocks (22.3) keep dependence shorter than a block: the volatility's interval widens 3.9 times, the Kelly fraction's 1.2 times.
- The book's block length, 145 for TR23, is the largest Politis–White estimate on the squared series.
- The crash gap's interval excludes 0. Near a bound the percentile interval stays in range, and the basic interval corrects a bias.
- A bootstrap test resamples under the null, (22.4).
- arch draws NumPy's block indices from the same generator; SciPy resamples days only.

PRACTICE PROBLEMS

Problems use TR23 ($T = 5{,}849$ daily simple returns from 3 January 2003, divisor T, $A = 246.12$) or U28 (price indices). Bootstrap standard errors divide by B; intervals are 95% percentile intervals unless stated; experiment `e` uses `default_rng([e, 20261011])`.

1. Three daily returns are +1%, −2% and +4%. The ideal bootstrap standard error of their mean, over all 27 resamples, is *closest* to:

   A. 1.414%

   B. 1.732%

   C. 0.816%

2. With $B = 10{,}000$ resampled values, the ends of a 95% percentile interval are:

   A. the 500th and 9,500th smallest values

   B. the 250th and 9,750th smallest values

   C. the estimate ± 1.96 bootstrap standard errors

3. In 2020 the first component's share is 66.62%; block resamples give the percentile interval 49.55% to 75.92% and the standard error 7.11 points. The 95% basic interval is *closest* to:

   A. 49.6% to 75.9%

   B. 52.7% to 80.6%

   C. 57.3% to 83.7%

4. iid resamples of TR23 put $\hat{\rho}_1(|r|)$ between −0.025 and 0.026, against the estimate 0.161. The best reading is:

   A. 0.161 is an outlier of the sample, since resamples rarely produce it

   B. the resamples describe independent days, so their range is a null distribution that 0.161 rejects

C. the bias is −0.161, so the basic interval 0.295 to 0.346 corrects it

5. In a stationary bootstrap with $\ell = 145$ on $T = 5{,}849$ days, the probability that a given position after the first starts a new block is *closest* to:

   A. 0.9931

   B. 0.0248

   C. 0.0069

6. A stationary bootstrap has $\ell = 145$ and $T = 5{,}849$. (a) Find the expected number of blocks in a resample. (b) Find the chance that a block lasts 60 days or more. (c) Estimate the share of adjacent resampled pairs that are adjacent in TR23, and predict the resamples' mean $\hat{\rho}_1(|r|)$ at $\ell = 145$ and $\ell = 2$. Experiment 3 gives 0.156, and experiment 21 ($\ell = 2$, $B = 2{,}000$) gives 0.078 ± 0.0005.

7. In TR23's 7% regime, stationary blocks with $\ell = 108$ (experiment 17, $B = 10{,}000$) give 477 resampled |t-stat| at or above the observed 2.05 under $\mu = 0$. (a) Find the two-sided bootstrap p-value and its Monte Carlo standard error. (b) Decide at the 5% level; why is the decision fragile? (c) Which B brings the Monte Carlo standard error to 0.001 at the estimated p-value?

8. In Example 9, why does the basic interval fail? Why do a third of the resamples have no window under water, and why is their mean standard deviation of $\ln G$ 0.386, not 0.283?

9. (a) Prove that every position of a stationary resample is uniform on the T days. (b) How many moving blocks of 145 days contain day 1, and how many day 145? (c) On TR23 a random moving block of 145 days has expected mean 0.060185%. Compare it with $\bar{R} = 0.062505\%$.

10. *(Python)* Write the stationary bootstrap's index function and check it against arch on a twin generator (experiment 20). Then reproduce Exhibit 1's volatility interval from experiment 3 ($\ell = 145$, $B = 10{,}000$).

11. *(Python, stability check)* For TR23's 10% regime (2,772 returns from 1 June 2015), compute $\hat{f}$, the book's block length, and iid (experiment 22) and block (experiment 23) intervals. Compare them with the full sample's 2.2 to 6.5 (iid) and 1.8 to 7.1 (blocks).

SOLUTIONS

1. **A** is correct. Under $\hat{F}$ a day has variance s^2 with divisor 3, so $s = 2.4495\%$, and by (18.8) the mean of three draws has standard error $s/\sqrt{3} = 1.414\%$. B, 1.732%, uses s_{u}, but $\hat{F}$ divides by T. C, 0.816%, is $s/3$, without the square root.

2. **B** is correct. The ends are the order statistics $\lceil 0.025B \rceil = 250$ and $\lceil 0.975B \rceil = 9{,}750$. A gives a 90% interval, with α not halved. C is a normal interval, not a percentile one.

3. **C** is correct. By (22.2) the basic interval runs from $2 \times 66.62 - 75.92 = 57.32$ to $2 \times 66.62 - 49.55 = 83.69$. A is the percentile interval, and B, $66.62 \pm 1.96 \times 7.11$, is a normal interval that ignores the skew.

4. **B** is correct. iid resamples have independent days by construction, so they show what independence would produce, and 0.161 lies far outside. A reads the resamples as a picture of TR23's dependence, which they destroy. C treats −0.161 as a bias, but a bias estimate means nothing for a quantity the resampling destroys.

5. **C** is correct. By (22.3) a new block starts with probability $1/\ell = 1/145 = 0.0069$. A, 0.9931, is the chance that the block continues. B, $\ell/T = 0.0248$, is the share of the sample that one average block covers.

6. (a) The first position starts a block, and each of the other 5,848 starts one with probability $1/145$: $1 + 5{,}848/145 = 41.3$.

   (b) By (22.3), $(1 - 1/145)^{59} = 0.665$.

   (c) A pair is adjacent in TR23 when its block continues, with probability $1 - 1/\ell$; a new block lands on the next day only with probability $1/T$. The other pairs carry almost no dependence, so the mean is about $(1 - 1/\ell) \times 0.161$: about 0.16 at $\ell = 145$ and 0.08 at $\ell = 2$, against 0.156 and 0.078.

   Each resample is centred on its own mean, which varies more under long blocks; this lowers the $\ell = 145$ value to about 0.156. At $\ell = 2$ the expectation is about 0.080 (experiment 26, $B = 200{,}000$: 0.0797 ± 0.00005), and experiment 21 lies about 3 Monte Carlo standard errors below it.

7. (a) The p-value is $477/10{,}000 = 0.0477$, with Monte Carlo standard error $\sqrt{0.0477 \times 0.9523/10{,}000} = 0.0021$.

   (b) Reject, since $0.0477 < 0.05$. But $0.0477 + 2 \times 0.0021 = 0.052$ exceeds 0.05, so another set of resamples, or another block length, could cross the line.

   (c) About 45,000, since $0.0477 \times 0.9523/0.001^2 = 45{,}425$.

8. The basic interval reflects the resampled shares about 4.14%, so its lower end, $2 \times 4.14\% - 33\% \approx -25\%$, is below 0, where no share can lie. The windows overlap and hold fewer than five disjoint five-year periods (LM18), so whether any window ends under water depends on one or two long falls; 3,248 resamples have none.

   Blocks cut TR23's swings of several years, which held the windows' spread at 0.283. iid days give this estimator 0.357, below 0.425 because the windows overlap; the swings within months that blocks keep lift the mean to 0.386.

9. (a) This is Deep Dive (b).

   (b) Of the 5,705 blocks, day 1 lies only in the one that starts on it, and day 145 in the 145 that start on days 1 to 145.

   (c) A random block weights each day by the number of blocks that contain it, so the first and last 144 days count less. Both stretches averaged more than $\bar{R}$ (0.13% and 0.28% a day), so the block mean is lower. The difference, $0.060185\% - 0.062505\% = -0.00232\%$, is −0.23 bp, −3.7% of $\bar{R}$ or 0.15 standard errors of $\bar{R}$.

10. Section 9's function and twin check print `True` three times. Then:

PYTHON

```python
rng = np.random.default_rng([3, 20261011])      # experiment 3
vol = np.empty(10_000)
for c in range(0, 10_000, 500):                 # chunks of 500 rows
    idx = np.stack([stationary_idx(rng, len(R), 145)
                    for _ in range(500)])
    vol[c:c + 500] = np.sqrt(A) * R[idx].std(axis=1)
print(np.quantile(vol, [0.025, 0.975], method="inverted_cdf"))
```

It prints 0.1648... and 0.2150..., the interval 16.5% to 21.5%.

11. In the 10% regime $\hat{f} = 6.06$, and the book's rule gives $\ell = 36$. iid days give 2.8 to 9.6 and blocks 2.6 to 10.1. They overlap the full sample's 2.2 to 6.5 (iid) and 1.8 to 7.1 (blocks) from 2.8 to 6.5 and from 2.6 to 7.1. The full sample includes the regime's 2,772 days, so the intervals are not independent. This is a stability check, not a test on later data.

GLOSSARY

Term	中文	Meaning
Sampling distribution	抽樣分布	The law of an estimate across samples drawn from F.
Bootstrap	自助重抽法	Reading an estimate's sampling distribution from resamples drawn from $\hat{F}$ (or from blocks).
Resampling	重抽樣	Drawing a new sample from the data.
Ideal bootstrap	—	The exact law of the resampled statistic, over all T^T equally likely resamples.
Bootstrap standard error	標準誤	The standard deviation of B resampled values, divisor B, (22.1).
Percentile interval	百分位數信賴區間	The $\frac{\alpha}{2}$ and $1-\frac{\alpha}{2}$ quantiles of the resampled values, (22.2).
Basic interval	基本信賴區間	$2\hat{\theta}$ minus the percentile interval's ends, reversed, (22.2).
BCa interval	偏誤加速校正之自助重抽信賴區間	The percentile interval adjusted for bias and skewness (Efron 1987).
Block bootstrap	區塊自助重抽法	Resampling stretches of consecutive days.
Moving blocks	移動區塊	Blocks of ℓ consecutive days drawn from the $T-\ell+1$ in the sample.
Circular blocks	循環區塊	Moving blocks on a circle, day T followed by day 1.
Stationary bootstrap	平穩自助重抽法	Blocks of random length with mean ℓ, (22.3).
Block length	區塊長度	ℓ; for TR23 the book's rule gives 145.
Bootstrap test	自助重抽檢定	A test whose null law comes from resamples of data altered so that the null hypothesis holds.
Bootstrap p-value	p值	The share of resamples under the null at least as extreme as the data, (22.4).

REFERENCES

- Efron, B. (1987). "Better Bootstrap Confidence Intervals." *Journal of the American Statistical Association* 82 (397): 171–185.
- Efron, B., and R. J. Tibshirani (1993). *An Introduction to the Bootstrap*. Chapman & Hall, ch. 4 (the plug-in principle), §6.2 and its equation 6.6 (the bootstrap standard error, divisor $B - 1$) and ch. 13 (percentile intervals).
- Künsch, H. R. (1989). "The Jackknife and the Bootstrap for General Stationary Observations." *Annals of Statistics* 17 (3): 1217–1241.
- Lahiri, S. N. (2003). *Resampling Methods for Dependent Data*. New York: Springer, §2.7.1 (circular blocks), Theorems 3.1 and 3.3 (block bootstraps of a mean) and §7.3 (subsampling).
- Ledoit, O., and M. Wolf (2008). "Robust Performance Hypothesis Testing with the Sharpe Ratio." *Journal of Empirical Finance* 15 (5): 850–859.
- Lo, A. W. (2002). "The Statistics of Sharpe Ratios." *Financial Analysts Journal* 58 (4): 36–52.
- Patton, A. J., D. N. Politis and H. White (2009). "Correction to 'Automatic Block-Length Selection for the Dependent Bootstrap' by D. Politis and H. White." *Econometric Reviews* 28 (4): 372–375.
- Politis, D. N., and J. P. Romano (1994). "The Stationary Bootstrap." *Journal of the American Statistical Association* 89 (428): 1303–1313.
- Politis, D. N., and H. White (2004). "Automatic Block-Length Selection for the Dependent Bootstrap." *Econometric Reviews* 23 (1): 53–70.
- Wasserman, L. (2004). *All of Statistics: A Concise Course in Statistical Inference*. New York: Springer, §9.11 (the parametric bootstrap).
- arch 8.0.0: `StationaryBootstrap`, `CircularBlockBootstrap`, `MovingBlockBootstrap` and `optimal_block_length`.
- SciPy 1.18 reference: `scipy.stats.bootstrap`, checked against SciPy 1.18.1; NumPy 2.5 reference: `numpy.quantile`, checked against NumPy 2.5.3.
- FinMind open data: `TaiwanStockTotalReturnIndex` (data_id TAIEX), retrieved 18 October 2026; `TaiwanStockPrice` (TWSE sector indices), retrieved 11 October 2026, with closes repaired from `TaiwanStockEvery5SecondsIndex` as in LM16.